

Jay Burgess

Principal Forward Deployed Engineer with 15+ years translating complex operational, regulatory, and customer problems into production software and data systems. Built this capability across frontline systems analysis, policy automation, enterprise data platforms, full-stack products, and agentic AI before Forward Deployed Engineer became a standardized title. Delivered 17+ production systems across regulated and Fortune 500 environments, including redesigning a core operational workflow into new production software that generated \$2.7M in new revenue within three months of rollout. Owns the complete delivery lifecycle from discovery and architecture through deployment, governance, adoption, and operational handoff.

EMAIL

jayburgessjr@gmail.com

PHONE

[213-949-4813](tel:213-949-4813)

LINKEDIN

linkedin.com/in/jayburgessjr

GITHUB

github.com/jayburgessjr

\$2.7M

NEW CLIENT REVENUE, 3
MONTHS

65%

FASTER CLIENT
ONBOARDING

50%

FASTER COMPLIANCE
REVIEW

Revuity Systems, a core operational workflow rebuild	Multi-tenant SaaS platforms, Revuity Systems	Stateful multi-agent workflows with human approval gates
86% FASTER EXECUTIVE REPORTING ~\$6B global marketing spend governed, Adobe	17+ PRODUCTION SYSTEMS DELIVERED Across regulated and Fortune 500 environments	

01 Experience

Fifteen years.
One thread: ship it, then hand it off.

● **Revuity Systems** Principal Forward Deployed Engineer

- Operate Revuity Systems as an independent forward deployed engineering practice, delivering 17+ production systems, 15+ reusable products, and five service offerings across regulated and enterprise environments. Redesigned a core operational workflow into new production software, generating \$2.7M in new revenue within three months of rollout.
- Led discovery through operational handoff with executives and operators, converting ambiguous workflows into solution architectures, prioritized backlogs, acceptance criteria, and production releases that enabled client teams to operate delivered systems independently.
- Architected multi-tenant SaaS platforms with Next.js, FastAPI, Supabase, PostgreSQL, pgvector, custom MCP servers, RBAC, row-level security, and audit logging, reducing onboarding time by 65%.
- Designed stateful multi-agent workflows with persisted checkpoints, asynchronous execution, deterministic routing, structured outputs, and human approval gates, reducing compliance review turnaround by 50%.
- Built Revuity Agentic OS with eight agents, 12 MCP servers, and 197 reusable skills, while platform consolidation and white-label deployments produced \$250K+ in SaaS savings and a 15% gain in marketing and sales performance.

- Engineered production LLM controls, including tool contracts, structured outputs, evaluations, deterministic routing, audit trails, and human approval, enabling traceable and governed AI-assisted decisions in regulated environments.

● **Higher Education Assistance Group (HEAG)**

Principal AI Systems Architect and Forward Deployed Engineer

- Architected and deployed a multi-agent compliance platform with Claude, LangGraph, and MCP, increasing case-processing capacity by 25% without added headcount.
- Engineered production LLM controls including tool contracts, structured outputs, evaluations, deterministic routing, audit trails, and human approval for regulated decision support.
- Shipped OpsFlow end to end with Claude Code, FastAPI, and Next.js to replace a misaligned Jira workflow and reduce operational reporting cycle time by 40%.
- Converted evolving Title IV policy into deployable software with executives, consultants, and frontline operators, creating a governed knowledge pipeline that preserved traceable review, approval, and accountability across regulated content updates.

● **Walmart**

Senior Data Scientist and Analytics Lead, Creative Operations

- Partnered with creative operations leaders to identify production bottlenecks and translate workflow problems into structured data models, analytics products, and decision-support tools.
- Built SQL and dbt pipelines that unified creative production and campaign operations data into governed models and real-time operating metrics.
- Developed Python and scikit-learn anomaly-detection models to surface spend spikes, workflow deviations, and production constraints, contributing to a 37% improvement in project completion rates.

● **Adobe**

Senior Analyst, Marketing Operations Analytics

- Worked across marketing, finance, and operations teams to translate enterprise reporting requirements into production data pipelines and standardized analytical models.

- Built Python, SQL, and PySpark pipelines across Airflow, Amazon S3, EMR, and Redshift to govern approximately \$6B in global marketing spend and reduce executive reporting turnaround by 86%.
- Strengthened production reliability through Docker, CI/CD, automated data-quality tests, schema validation, monitoring, and failure alerting, reducing manual deployment risk and improving the consistency of enterprise reporting pipelines.



P3 Cost Analysts

Senior Data Systems Engineer, Revenue Operations

- Embedded predictive retention intelligence into daily operations to improve pricing decisions, client-risk visibility, and portfolio forecasting.
- Built the platform end to end using a Next.js front end, custom backend services and APIs, PostgreSQL and SQL Server data models, and Power BI reporting.
- Integrated a machine-learning-driven customer-retention model into the platform, improving the organization's ability to identify account risk, evaluate pricing decisions, and forecast portfolio revenue.



Araya Rose Analytics

Senior Data Engineer

- Worked directly with client leaders to translate revenue and reporting challenges into CRM, analytics, and executive decision platforms, replacing fragmented information with more consistent access to operational and performance data.
- Created reusable data models and governed workflows that replaced manual reporting with repeatable processes, standardized business logic, and maintainable reporting infrastructure.
- Delivered client-facing systems from requirements discovery through implementation and adoption, giving leadership teams more consistent access to revenue, customer, and performance information.



Columbia College Hollywood

Assistant Director, Financial Aid Systems

- Partnered with financial aid leaders and frontline staff to identify compliance risks, workflow bottlenecks, and reporting gaps across Title IV administration.

- Automated federal financial aid workflows and built compliance dashboards that converted policy requirements into repeatable operational controls and review processes.
- Improved the consistency and auditability of financial aid operations by centralizing validation, exception monitoring, and management reporting.



Loyola Marymount University

Systems Analyst and Financial Aid Counselor

- Combined frontline financial aid counseling with systems analysis to uncover workflow gaps and translate federal policy into reporting, validation, and technical controls, improving the reliability of regulated financial aid operations.
- Built regulatory reporting and data-validation processes that helped staff identify incomplete records, resolve exceptions, and support compliant Title IV administration.
- Established the domain and user-centered foundation for later FDE work by solving operational problems beside the staff responsible for executing regulated processes.

02 Technical Skills

Full stack.

Discovery to production.

AGENTIC AI & GOVERNANCE

Claude OpenAI LangGraph Model Context Protocol (MCP) Multi-agent orchestration RAG

Tool calling Structured outputs Evaluation harnesses Human-in-the-loop review

Deterministic guardrails Audit logging

SOFTWARE & DATA ENGINEERING

Python TypeScript JavaScript SQL FastAPI asyncio Node.js Next.js React

REST APIs PostgreSQL pgvector Supabase SQL Server dbt PySpark Apache Airflow

CLOUD, SECURITY & DELIVERY

AWS Amazon S3 EMR Redshift Microsoft Azure Docker CI/CD GitHub Actions

GitLab

RBAC

Row-level security

Monitoring

Automated validation

FORWARD DEPLOYED PRACTICE

Technical discovery

Stakeholder workshops

Requirements decomposition

Solution architecture

Rapid prototyping

Production deployment

Runbooks

Architecture decision records

Adoption

Team enablement

03

Publications, Certifications & Education

Written it down.
Credentialed it too.

PUBLICATIONS

[Jay's Art of Systems Thinking](#)

[The Forward Deployed Engineer: A Simple Guide to Building What Enterprises Actually Need](#)

[Agentic Engineering: Building Autonomous AI Systems That Actually Work](#)

[The Agentic Builder: Field Notes on Product Development](#)

[Stories from a Systems Thinker: Twenty-Five Parables on Why the Obvious Fix Never Works](#)

CERTIFICATIONS

Certified Forward Deployed Engineer (CFDE)

Certified Agentic Engineer (CAE)

Claude Partner Badge, Claude Code, Anthropic

IBM AI Engineering Professional Certificate

Microsoft Certified: Azure AI Fundamentals

EDUCATION

Master of Business Administration (MBA)

Eastern University · 2024

Master of Science, Data Science

Eastern University · 2022

Master of Education (M.Ed.)

Loyola Marymount University · 2013

Bachelor of Arts, Sociology

Loyola Marymount University · 2005